Lekh Nayak

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Education

Sardar Patel Institute Of Technology **Aug 2023 – Present**
B.Tech in Computer Science Engineering *CGPA: 8.2/10*

IIT Patna **Mar 2025 – Present**
Minor in Business Analytics *CGPA: 9.4/10*

Experience

Cactus Life Sciences **Jun 2026 – Aug 2026**
SDE Intern *Remote*

- Developed a **SaaS platform** combining **LLM-powered text, infographic, and presentation generation** through **multi-agent AI workflows** and automation pipelines.
- Secured the platform with **Microsoft OAuth 2.0/OIDC**, containerized services with **Docker**, deployed on **Azure Static Web Apps**, and integrated **Azure Blob Storage** for file and PDF hosting.

Sardar Patel Institute of Technology **Dec 2025 – Feb 2026**
System Engineering Intern – GPU Access Portal *Mumbai*

- Engineered a scalable **Linux HPC cluster** using **Slurm**, **NFS**, and **SSSD/LDAP**, integrating **Prometheus** and **Grafana** for real-time monitoring of GPU usage, job queues, and node health.
- Deployed **Open OnDemand** for browser-based access to **Jupyter**, **VS Code Server**, and **RStudio**, eliminating local CLI setup for end users.

Projects

InterviewMate | *React, Node.js, Express, MongoDB, Flask, Groq (Llama-3.3 / Whisper), ChromaDB* **Sept 2025 - Dec 2025**

- Architected a **real-time voice-based mock-interview platform** on a **microservices architecture** (React + Node/Express + Python Flask), integrating **Groq Llama-3.3-70B** for adaptive question generation, **Whisper** for speech-to-text, and **Sarvam TTS** to conduct spoken interviews across **5 configurable interview modes** (HR, DSA, technical, project, hybrid).
- Built a **semantic resume-to-JD matcher** with **SentenceTransformer embeddings** and a **ChromaDB** vector store, generating **cosine-similarity ATS scores** and LLM-driven skill-gap reports.
- Designed **session storage and analytics** in MongoDB (JWT + OTP auth) with **4-dimensional feedback scoring** and a **Keras speech-emotion model** for vocal-delivery cues.

Collaborative VS Code | *React, Node.js, Express, WebSocket, Monaco Editor, Docker, PostgreSQL* **Jan 2026 - Mar 2026**

- Engineered a **real-time collaborative code editor** stress-tested with **30+ concurrent users**, maintaining **synchronized cursors, terminals, and file states** with under **100ms sync latency**.
- Integrated **Monaco Editor** with syntax highlighting and intelligent suggestions across **6+ languages**, executing code in **Docker-isolated sandboxed environments** for safe multi-user runs.
- Designed a **WebSocket-driven event system** handling real-time collaboration, code execution broadcasts, and **state consistency** across all connected clients with zero data loss on reconnection.

CtrlFixIt | *Next.js, TypeScript, PostgreSQL, Redis, Docker, WebSockets, Kafka* **Jan 2026 - Feb 2026**

- Built a **production-grade competitive coding platform** deployed live for **300+ participants**, handling **150+ concurrent sandboxed code execution jobs** via a Piston-based Docker engine clustered with Kafka and tunneled through ngrok.
- Engineered a **real-time event system** using WebSockets for live scoreboards, instant code execution feedback, and **server-authoritative timing** to ensure fairness across all connected clients.
- Optimized for **vertical scalability** using Redis-backed queuing and Prisma ORM with PostgreSQL, supporting multi-language execution (Python, C++, Java, Go, Rust, JavaScript) with **1GB isolated temp memory** per job.

Technical Skills

Languages: JavaScript, TypeScript, SQL, Java, Python
Frontend: HTML5, CSS3, React, Next.js
Backend: Node.js, Express.js, RESTful APIs, WebSockets, OAuth 2.0/OIDC, JWT, Kafka
Databases: PostgreSQL, MongoDB, Redis, Prisma ORM, ChromaDB
DevOps & Tools: Git, Docker, Linux, Azure, Prometheus, Grafana
AI/ML: PyTorch, Scikit-learn, NLTK, BERT, LLM APIs

Position of Responsibility

Computer Society of India **Aug 2025 – May 2026**
Head of Events *Sardar Patel Institute of Technology*

- Led chapter operations by managing a 60+ member team, driving strategic initiatives, and organizing technical events for 350+ participants through effective stakeholder coordination and end-to-end execution.

Extra Curricular & Achievements

- **1st Place:** La Casa De Code - National Level Hackathon (Dayananda Sagar University), 2x Abhimat - Lok Sabha Style Debate, CSI S.P.I.T ML Competition (Sardar Patel Institute of Technology, Mumbai)
- **2nd Place:** EFOS Virtual Hackathon - EFOS, Runner up Best Delegate - AIPPM (mUNIFY S.P.I.T. Mumbai), MergeMania - Mergers & Acquisition (E-Summit 2025 S.P.I.T. Mumbai), Dreams and Deals - Pitching Contest, Pitch Hunt - Elevator Pitch Challenge, Annual Oxford Style Debate (S.P.C.E. Mumbai)
- **Finalist:** HERE Geospatial Hackathon (Pan India Hiring Hack), Minithon (IIIT Bhubaneswar – top 9/60+), Airavat AI Hackathon (SPIT Mumbai 2024), Inspiriobiz 5.0 (VESIM Mumbai – top 10/300+ teams), Vishleshan – Canfem Case Study (TISS Mumbai – top 5/550+ teams), BizQuest (E-Summit 2024 SPIT Mumbai)